

On the Categorical Propagation of Contact in Living Systems: Catalysis as Categorical Provision Upon Contact

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Abstract

Catalysis is conventionally described but not derived. Transition-state theory tells us that a catalyst lowers an activation barrier; it does not tell us why such objects exist, why they are reusable, why they are specific, or why specificity is bounded above as well as below. We derive catalysis from two primitives. The first is that individuation is not free: distinguishing an item from its complement takes finite time, during which the complement evolves, so every boundary carries irreducible thickness $\beta > 0$. The second is that a *category* is the unit of a step—the structure that permits an invariant to persist across a transition in which nothing numerically survives. We prove these primitives are independent: the first drives systems toward configurations of minimum boundary cost, but it cannot account for reuse, because the uncertainty a process resolves is consumed in resolving it. Reuse requires something that is identically the same across occasions while each occasion is a distinct event, and that is what a category is and residue is not. From this we obtain the definition: *a catalyst is an item that, upon contact, increases the number of completable categories available to a transition, and is not consumed because categorical provision requires no free energy*. Nine results follow as corollaries rather than assumptions: catalysts are not consumed; equilibrium constants are unaltered; the Haldane relation is recovered; rate

enhancement is exponential in the number of apertures traversed with a single framework-wide constant; specificity is bounded on both sides by the joint requirement of provision and release, giving a *specificity window*; competitive inhibition is contact without provision; mechanism-based inactivation is provision without release; allosteric regulation is modification of the provider’s contact chain; and catalysis is not a Maxwell demon, because categories are configurational and the demon’s criterion is kinetic, quantities which depend on disjoint variables. Because the account never requires that the provider be a protein, it resolves the origin-of-life regress in which enzymes require proteins and proteins require nucleic acid templates: the first catalysts were structures whose delocalised electronic motifs supplied categories on contact and were unchanged by their use. We give an abstract system exhibiting the same mechanics with no chemistry in it, establishing that the categorical primitive is not a chemical postulate. All claims are stated as falsifiable predictions and are accompanied by computational validation against public reaction and kinetics databases.

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Part I

The Problem

1 Introduction

1.1 What is not explained by the standard account

The standard account of enzyme catalysis is transition-state stabilisation [Pauling, 1946, Eyring, 1935]. An enzyme binds the transition state more tightly than the ground state; the activation free energy falls; the rate rises [Schramm, 2011, Warshel et al., 2006]. This account is correct as far as it goes, and it is quantitatively successful. It is also, as an explanation, incomplete in five specific ways, and the incompleteness is not a matter of missing detail but of missing derivation.

First: why does anything play this role at all? Transition-state stabilisation describes what an enzyme does once we are given an enzyme. It does not explain why objects of this kind should exist, or why matter organised into cells should invariably produce them. To say that enzymes exist because they are useful is to give a selective rather than a physical answer, and it presupposes the existence of the entities selection acts upon.

Second: why is the catalyst reusable?

This is the question we shall find to be decisive. A catalyst participates in a reaction and emerges unaltered, ready to participate again. On any account in which catalysis consists in the resolution of some uncertainty, or the traversal of some barrier, or the payment of some cost, the first turnover ought to exhaust the thing that made it possible. That it does not is normally recorded as a definitional stipulation—"a catalyst is not consumed"—rather than derived. We shall show that the requirement of reuse is strong enough to force a second primitive into the theory.

Third: why is a catalyst specific? Specificity is usually explained by complementarity [Fischer, 1894, Koshland, 1958]: the substrate

fits the site. But complementarity explains only the lower bound—why a badly matched substrate is not turned over. It does not explain the upper bound. Nothing in the lock-and-key picture forbids a lock so exquisitely matched to its key that the two never separate. Yet such objects are not enzymes; they are, at best, single-use reagents. Real enzymes occupy a bounded region of specificity, and the bound above is not explained.

Fourth: why do inhibitors fall into the classes they do? Competitive inhibitors occupy the site and do nothing. Mechanism-based inactivators are turned over once and destroy the enzyme in the process [Abeles and Maycock, 1976, Silverman, 1995]. These are treated as separate pharmacological phenomena. On the account developed here they are the two ways the catalytic requirement can fail, and they are therefore predictions rather than observations.

Fifth: what were the first catalysts?

Life requires enzymes. Enzymes, in extant biology, are proteins. Proteins require templated synthesis, hence nucleic acids, hence a replication apparatus that is itself enzyme-dependent [Eigen, 1971, Gilbert, 1986, Orgel, 2004]. The regress is well known and the standard resolutions—ribozymes [Kruger et al., 1982, Guerrier-Takada et al., 1983], surface metabolism [Wachtershauser, 1988], autocatalytic sets [Kauffman, 1986]—all proceed by exhibiting a non-protein catalyst. What is missing is a criterion that tells us what makes something a catalyst independently of what it is made of. Without such a criterion the resolutions are existence proofs without a theory.

1.2 What this paper does

We derive catalysis from two primitives, prove they are independent, and obtain the five missing explanations as consequences.

The first primitive (section 2) is that individuation costs something irreducibly. To be a distinct item is to be distinguished from everything one is not; establishing that distinction takes non-zero time; during that time

the complement changes; so the boundary is not a clean cut but a layer of finite thickness $\beta > 0$. We derive this rather than assume it, and we derive it in a setting where the bound is uniform, which is the technically delicate point.

The second primitive (section 3) is the *category*. We motivate it by an argument from reuse: what a catalyst must have, in order to be usable twice, is an invariant that is numerically the same on both occasions while the occasions are genuinely distinct events. We show that the first primitive cannot supply this, and that the structure which can is what we shall call a category. We then give the measure—categorical entropy—and prove the property that makes catalysis possible: completing a category increases entropy without consuming free energy.

Section 4 exhibits an abstract system, containing no chemistry whatever, which has both primitives and in which the second is demonstrably not reducible to the first. This is included because the central claim of the paper is a claim about what kind of thing a catalyst is, and such a claim should not depend on the accident that our examples are molecules.

Part III contains the definition and its consequences. Part IV states the falsifiable predictions and the validation protocol. Part V applies the definition to the origin-of-life regress.

1.3 Conventions

Throughout, $\mathcal{U}$ denotes the totality under discussion and $\mathcal{I}$ an item individuated within it. We write $\beta(A, B)$ for the boundary thickness between A and B and $\beta_{\min}$ for its lower bound. k_B is Boltzmann’s constant. Logarithms are natural unless subscripted. We use Ω for a count of microstates in the Boltzmann sense [Boltzmann, 1877, Gibbs, 1902] and $|\gamma(t)|$ for a count of completed categories. Standard enzyme-kinetic notation follows Cornish-Bowden [1979]: k_{cat} , K_M , K_{eq} , with k_{cat}/K_M the specificity constant.

We are careful throughout to distinguish results that are unconditional from those that hold under stated hypotheses, and we mark the latter explicitly.

Part II Foundations

2 Cost of Individuation

2.1 Existence by distinction

Physics customarily begins with objects already individuated: particles, systems, states. The process by which something comes to be a distinct something is not usually part of the formalism. We make it part of the formalism, because catalysis turns out to be a fact about that process.

Postulate 1 (Existence by distinction). An item $\mathcal{I}$ exists as a distinct entity if and only if there is a well-defined complement $\mathcal{U} \setminus \mathcal{I}$ with $\mathcal{I} \cap (\mathcal{U} \setminus \mathcal{I}) = \emptyset$ and $\mathcal{I} \cup (\mathcal{U} \setminus \mathcal{I}) = \mathcal{U}$. The existence of $\mathcal{I}$ consists in the distinction $\mathcal{I} \neq \mathcal{U} \setminus \mathcal{I}$, and not in any property $\mathcal{I}$ possesses independently of it.

This is not a metaphysical flourish. It is the observation that every physical predicate we assign to an item—its mass, its charge, its position—is assigned by contrast with what does not have it. Postulate 1 says only that we should take that contrast as constitutive rather than incidental.

Postulate 2 (Distinction takes time). Establishing the distinction $\mathcal{I} \neq \mathcal{U} \setminus \mathcal{I}$ requires a finite, non-zero interval $\tau_p > 0$. During τ_p the complement $\mathcal{U} \setminus \mathcal{I}$ evolves.

Remark 2.1. Postulate 2 is forced rather than chosen. Suppose $\tau_p = 0$. Then the state of $\mathcal{U}$ immediately before the distinction and immediately after are the same state. But if nothing about $\mathcal{U}$ differs, no distinction has been drawn—the supposed individuation made no difference, and an individuation that makes no difference is not one. Hence $\tau_p > 0$. The second clause is then automatic for any $\mathcal{U}$ that is not static.

2.2 The boundary has thickness

Definition 2.2 (Boundary thickness). Let A be individuated within $\mathcal{U}$ over the interval $[t_0, t_0 + \tau_p]$. The *boundary thickness* $\beta(A)$ is

the measure of the set of states which, at the completion of the individuation, belong determinately neither to A nor to $\mathcal{U} \setminus A$.

Theorem 2.3 (Positivity of thickness). *For any individuation carried out by a process satisfying Postulates 1 and 2, $\beta(A) > 0$.*

Proof. Let the individuation begin at t_0 and complete at $t_0 + \tau_p$ with $\tau_p > 0$ by Postulate 2. Consider a state s that lies in $\mathcal{U} \setminus A$ at t_0 and in the region assigned to A at $t_0 + \tau_p$, or conversely. Such states exist whenever $\mathcal{U}$ is non-static on the timescale τ_p , which Postulate 2 guarantees. For such s the assignment made at the start of the operation and the assignment appropriate at its completion disagree, so s is not determinately in A nor determinately in $\mathcal{U} \setminus A$. The set of such states is non-empty, and $\beta(A)$ is its measure, hence positive. $\square$

Theorem 2.3 gives positivity for each individuation separately. For the results of part III we require something stronger: a bound that is uniform over the individuations a given system can perform. Positivity of each member of a family does not imply positivity of the infimum—the numbers $1/n$ are all positive and have infimum zero—so the uniform statement requires a hypothesis that bounds refinement.

Definition 2.4 (Resolution-bounded system). A system is *resolution-bounded* if the individuations it can perform are drawn from a finite set, or more generally if there is a finest individuation it can carry out, of thickness $\mu_{\min} > 0$.

Theorem 2.5 (Uniform floor). *In a resolution-bounded system there exists $\beta_{\min} > 0$ such that $\beta(A) \geq \beta_{\min}$ for every individuated A . One may take $\beta_{\min} = \mu_{\min}$.*

Proof. By theorem 2.4 the set of attainable thicknesses has a least element $\mu_{\min}$, or is bounded below by one. By theorem 2.3 every attainable thickness is positive, so $\mu_{\min} > 0$. The minimum of a finite set of positive numbers is positive; where the set is infinite the hypothesis of theorem 2.4 supplies the bound directly. Hence $\beta(A) \geq \mu_{\min} = \beta_{\min} > 0$ for all individuated A . $\square$

Remark 2.6 (Where the hypothesis is needed and where it is not). Theorem 2.3 is unconditional. Theorem 2.5 is conditional on theorem 2.4, and the condition is not cosmetic: a system able to refine without limit can have thicknesses approaching zero, and for such a system no uniform floor exists. Every physical system we shall consider is resolution-bounded, because the number of distinguishable cells in a bounded region of phase space is finite. We flag the dependence because results downstream inherit it.

2.3 Contact

Definition 2.7 (Contact). Items A and B are in *contact* when their mutual boundary thickness attains the floor:

$$\beta(A, B) = \beta_{\min}. \quad (1)$$

Contact is the closest two items can come while remaining distinct. Below $\beta_{\min}$ they are not two items; at $\beta_{\min}$ they are two items maximally mutually defined.

Proposition 2.8 (Local distinction is cheaper than global). *Maintaining A as distinct from a specific finite partner B requires boundary content bounded by $\beta(A, B)$; maintaining A as distinct from the whole of $\mathcal{U}$ requires boundary content bounded below by $\beta(A, \mathcal{U})$, and $\beta(A, B) \leq \beta(A, \mathcal{U})$ whenever $B \subseteq \mathcal{U} \setminus A$.*

Proof. The distinction of A from $\mathcal{U} \setminus A$ entails its distinction from every $B \subseteq \mathcal{U} \setminus A$, so the set of indeterminate states at the A – B boundary is a subset of that at the A – $\mathcal{U}$ boundary. Measures respect inclusion. $\square$

Corollary 2.9 (Contact is favoured). *Absent constraints preventing it, items evolve toward contact, because contact minimises the boundary content required to maintain mutual distinctness.*

Theorem 2.9 is a tendency, not a force. Nothing pulls the items together. Configurations of lower total boundary cost are simply the ones that persist, in the same sense in which a system samples its accessible states.

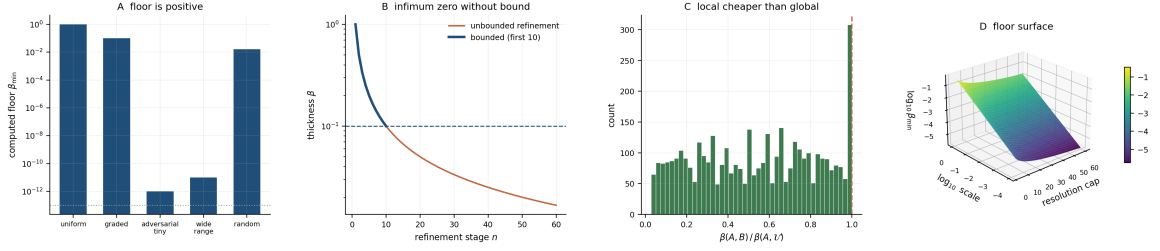


Figure 1: Individuation is not free: positivity of the boundary thickness and the hypothesis its uniformity requires. **(A)** Contact floor $\beta_{\min}$ computed—not assumed—as the minimum edge weight over five resolution-bounded structures spanning twelve orders of magnitude, including an adversarial instance whose smallest weight is 10^{-12} . Every instance returns a strictly positive floor; the dotted reference marks 10^{-13} . Positivity does not depend on the scale of the weights. **(B)** Thickness against refinement stage for a system permitted to refine without limit (orange), generating the sequence $1/n$. Each term is strictly positive and the sequence has infimum zero: every ε in a ladder down to 10^{-9} is eventually cleared. Truncating the same sequence to a resolution-bounded system (blue, first ten stages) restores a positive floor at 0.1 (dashed). Uniformity of the floor therefore requires the resolution-boundedness hypothesis and is not unconditional. **(C)** Distribution of the ratio $\beta(A,B)/\beta(A,U)$ over 4000 randomly generated nested structures. The ratio never exceeds unity (dashed), confirming that distinguishing an item from a specific finite partner costs no more than distinguishing it from the totality. Contact is the cheaper way to be distinct. **(D)** Floor surface $\log_{10} \beta_{\min}$ over the resolution cap and the weight scale. The surface is strictly negative-valued but nowhere reaches $-\infty$: for any finite cap the floor is bounded away from zero, and it descends toward zero only as the cap is removed. The two controlling variables act independently, the cap setting how finely the system may divide and the scale setting the size of the smallest available distinction.

Definition 2.10 (Contact chain). A *contact chain* is a sequence $A_1 - A_2 - \dots - A_n$ in which each consecutive pair is in contact.

Proposition 2.11 (Chains sharpen without global cost). *Let $A_1 - \dots - A_n$ be a contact chain and let A_0 come into contact with A_1 . Then each of $A_1, \dots, A_n$ is thereby distinguished more finely—each acquires an additional specific partner against which it is defined—while no $\beta(A_i, \mathcal{U})$ increases.*

Proof. The new contact establishes $\beta(A_0, A_1) = \beta_{\min}$ without altering any existing $\beta(A_i, A_{i+1})$, each of which is already at $\beta_{\min}$ by theorem 2.10. The context of each A_i now includes one further specific partner, which refines the distinctions available to it. No global boundary is modified, since A_0 was already individuated within $\mathcal{U}$. $\square$

2.4 What the first primitive gives, and what it does not

Postulate 2 has a consequence beyond thickness. Because individuation takes time and the complement moves, each individuation leaves an *unresolved remainder*: content that was one thing when the operation began and another when it completed, and which the operation therefore failed to assign. Call this the residue.

Proposition 2.12 (Residue drives boundary minimisation). *Accumulated residue is monotone: no operation removes it, since removal would require re-assigning content whose assignment was indeterminate, which is itself an individuation generating fresh residue. Systems therefore evolve irreversibly, and theorem 2.9 gives the direction: toward configurations of lower total boundary cost.*

This is a genuine dynamical principle, and it is all the first primitive gives. It is not enough, and the way in which it fails is the hinge of this paper.

Proposition 2.13 (Residue cannot account for reuse). *Let X be an item whose contribution to a transition consists solely in the residue that transition generates. Then X cannot support a second transition of the same kind.*

Proof. Residue is generated by the resolution of an indeterminacy. Once the transition has occurred, the indeterminacy it resolved is resolved: by theorem 2.12 it is not restored, since restoration is itself an individuation and generates its own residue rather than undoing the prior one. The condition under which X made the first transition possible therefore does not obtain at the second occasion. Hence X is, in the relevant respect, consumed. $\square$

Theorem 2.13 is the crux. A catalyst is by observation reusable. If everything a catalyst supplies were residue, catalysts would be single-use, and the phenomenon we are trying to explain would not exist. Something else must be supplied—something which is numerically the same on the second occasion as on the first, while the two occasions are distinct events. We now say what that is.

3 Category: The Unit of a Step

3.1 The requirement

Consider what theorem 2.13 demands. We require a structure $\mathfrak{C}$ with two properties that are in apparent tension:

- (i) **Invariance.** $\mathfrak{C}$ is the same at the second use as at the first. Were it different, what made the first turnover possible would not be what makes the second possible, and the reuse would be a coincidence rather than a mechanism.
- (ii) **Difference.** The first use and the second use are distinct events. Were they not distinct, no second turnover has occurred.

An item cannot be numerically identical across a change in which it participates and also be what changed. But it can be the *same category* while the *completions* of that category are distinct. This is the ordinary structure of a type and its tokens, imported into physics as a structural rather than a linguistic fact.

Definition 3.1 (Category). A *category* is a distinction that can be completed. It is specified independently of any particular completion; a *completion* is an occasion on which the distinction is drawn. The category is invariant across its completions; the completions are distinct events.

Remark 3.2 (Why this is a second primitive). A category is not constructed from residue and cannot be. Residue is what a completion leaves behind and is consumed in the leaving; a category is what makes completions of a given kind possible and survives them. The two are related as a capacity is related to its exercises. Theorem 2.13 shows the capacity is not recoverable from the exercises. We therefore take theorem 3.1 as primitive alongside Postulates 1 and 2, and section 4 exhibits a system in which the independence can be checked directly.

Remark 3.3 (Countability). Completions are countable: at any time a finite number have occurred, and $|\gamma(t)|$ denotes that number. The categories available for completion are not assumed countable, and nothing below requires that they be. All measures are defined on completions.

3.2 Categorical entropy

Definition 3.4 (Categorical entropy). For a system whose completed categories at time t number $|\gamma(t)|$, the *categorical entropy* is

$$S_{\text{cat}}(t) = k_B \ln |\gamma(t)|. \quad (2)$$

Definition 3.5 (Kinetic entropy). The *kinetic entropy* is the familiar quantity measuring the distribution of energy over accessible microstates,

$$S_{\text{kin}} = -k_B \sum_i p_i \ln p_i. \quad (3)$$

Theorem 3.6 (Completion increases entropy). Let $\Omega(C)$ be the number of microstates compatible with C completed categories. Then Ω is strictly increasing in C , and consequently

$$\frac{dS_{\text{cat}}}{dC} = k_B \frac{1}{\Omega(C)} \frac{d\Omega}{dC} > 0. \quad (4)$$

Proof. A completed category is a distinction the system did not previously draw: a new respect in which its configurations differ from one another. Configurations indistinguishable before the completion are distinguishable after it. Hence every microstate accessible at C remains accessible at $C + \Delta C$, and additional microstates—those differing only in the newly drawn distinction—become accessible. So $\Omega(C + \Delta C) > \Omega(C)$ for $\Delta C > 0$. Applying $S = k_B \ln \Omega$ and differentiating gives eq. (4). $\square$

3.3 The property that makes catalysis possible

Theorem 3.7 (Completion at zero free energy). *Categorical completion does not require free energy. There exist transitions with $\Delta F = 0$, $Q = 0$ and $W = 0$ which nevertheless complete a category and increase S_{cat} by at least $k_B \ln 2$.*

Proof. Consider a system at thermal equilibrium at temperature T , so that $\Delta F = 0$ for every spontaneous process. Let its internal configuration be specified by a vector of mode occupations $\mathbf{v} = (n_1, \dots, n_M)$, and consider a transition $\mathbf{v} \rightarrow \mathbf{v}'$ that redistributes occupation among modes at fixed total. By equipartition the mean energy per mode is fixed by T , so $\langle E(\mathbf{v}) \rangle = \langle E(\mathbf{v}') \rangle$: no net energy flows and $Q = 0$. No macroscopic displacement occurs, so $W = 0$. Hence

$$\Delta F = \Delta U - T\Delta S = 0. \quad (5)$$

But $\mathbf{v}$ and $\mathbf{v}'$ are distinguishable configurations, so the transition completes a category not previously completed. By theorem 3.6 this increases S_{cat} , and the minimal increase associated with drawing one binary distinction is $k_B \ln 2$. $\square$

Corollary 3.8 (Provision is free). *An item which supplies completable categories to another item, without itself undergoing a transition that consumes free energy, incurs no free-energy cost in doing so, and may repeat the provision indefinitely.*

Proof. Immediate from theorem 3.7: the completions the item enables are $\Delta F = 0$ transitions, and by theorem 3.1 the category itself is invariant across its completions. Nothing about the provider is spent. $\square$

Theorem 3.8 is the resolution of theorem 2.13. Residue is consumed; categories are not. An item that supplies categories can supply them again.

3.4 Rest: the limiting case

The argument from reuse (theorem 2.13) shows that residue is insufficient for a particular phenomenon. There is a second and sharper argument, which needs no catalyst at all, and which exhibits the two primitives separated at the limit.

Consider an item participating in no process: at rest, in its ground state, generating no residue whatever. Such an item nevertheless persists, and its persistence has structure. At each instant it is *this* item and not another—it is uniquely itself. Across instants it is the *same* item—it is invariant. These two requirements are the requirements of section 3, and they are in force even though nothing is happening.

Proposition 3.9 (Rest requires categorical structure). *Let X be an item generating no residue over an interval $[t_1, t_2]$. Then X persists over that interval only if it possesses a structure that is invariant across $[t_1, t_2]$ while being uniquely instantiated at each instant in it. Such a structure is a category in the sense of theorem 3.1, and it is not residue.*

Proof. Persistence of X over $[t_1, t_2]$ requires that X at t_2 be the same item as X at t_1 ; otherwise X did not persist but was replaced. It also requires that X at each $t \in [t_1, t_2]$ be determinately individuated, since by Postulate 1 to be an item at all is to be distinguished from one's complement, and that distinction obtains at each instant or not at all. The first requirement is invariance, the second is instantiation at each occasion; together they are theorem 3.1. That the structure is not residue is immediate: by hypothesis X gener-

ates none over the interval, so no residue is available to constitute it. $\square$

Corollary 3.10 (Persistence is not inertness). *An item persists into the future only in so far as its categories continue to be completed. The ground state is not the absence of categorical activity but categorical completion at zero residue.*

Proof. By theorem 3.9 the item's identity over an interval consists in a category instantiated at each instant of it. An instantiation is a completion (theorem 3.1). Hence persistence over $[t_1, t_2]$ is a succession of completions, one per instant, of an invariant category. By theorem 3.7 such completions need consume no free energy, and by hypothesis they generate no residue. $\square$

Remark 3.11. Theorem 3.10 inverts the usual reading of a ground state. It is customarily understood as the state in which nothing occurs. On the present account it is the state in which what occurs generates no residue—the categorical structure is fully engaged, and it is that engagement, not the absence of it, which constitutes the item's continuing to be there.

3.5 Independence of the two primitives

We record the logical situation, since it is the paper's structural claim.

Proposition 3.12 (The primitives are independent). *Neither primitive entails the other.*

- (a) *Thickness does not entail category. Theorem 2.13 exhibits a system—one in which all contribution is residue—which satisfies Postulates 1 and 2 and in which no reusable structure exists.*
- (b) *Category does not entail thickness. Section 4 exhibits a system with completable categories and reuse in which no individuation interval and hence no boundary thickness is defined.*
- (c) *The two are separated at the limit. Theorem 3.9 exhibits a case in which residue*

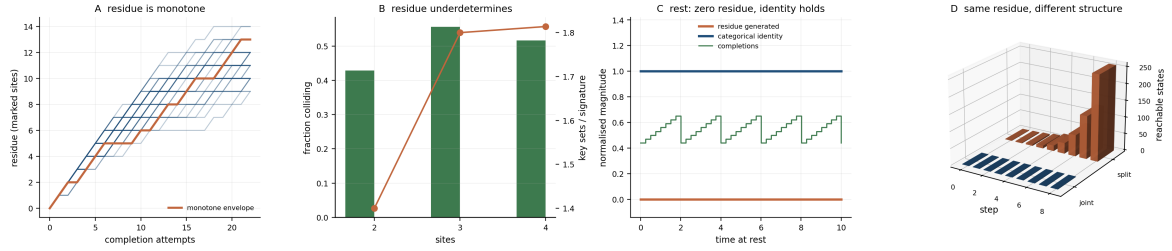


Figure 2: **Category and residue are independent primitives, and the independence is exhibited at zero residue.** (A) Forty independent trajectories of the abstract system, each attempting completions at randomly chosen sites over fourteen sites. Residue, measured as the number of marked sites, is non-decreasing along every trajectory: attempts on already-marked sites are no-ops and no operation unmarks. The monotone envelope is shown in orange. Irreversibility is a property of the accumulation, not of any individual step. (B) Exhaustive enumeration over all key sets of size at most three on two to four sites. Green bars give the fraction of key sets that share a terminal residue signature with at least one other key set; the orange curve gives the mean number of distinct key sets per signature. Distinct categorical structures routinely leave identical residue, so a record of what was resolved does not determine the capacities that did the resolving. (C) The limiting case. For a configuration admitting no applicable key, residue generated is identically zero (orange) for all time, while the categorical identity is invariant and fully determined (blue) and completions continue (green). A quantity that vanishes where another is fully engaged cannot be that other quantity. This is the separation of the two primitives at zero residue, and it requires no process to occur. (D) Reachable-state growth for two systems with identical terminal residue but different categorical structure: a single joint key that marks two sites at once, and two split keys that mark one site each. The split structure reaches exponentially more intermediate states. Terminal residue is blind to this difference; the reachable set is not.

is identically zero while categorical structure is fully in force. A quantity that vanishes where another is fully engaged cannot be that other quantity, nor be constructed from it.

Clause (b) is established constructively in the next section; clause (c) is established by theorem 3.9 and is, of the three, the strongest form of the separation, since it does not depend on any process occurring at all.

4 An Abstract System With the Same Mechanics

The claim that a category is a distinct primitive should not depend on the examples being chemical. We therefore exhibit a system with no physics in it, possessing both structures, in which their independence is a matter of inspection.

4.1 The system

Let V be a finite set of *sites* and let Σ be a finite set of *marks*. A *configuration* is a map $c : V \rightarrow \Sigma \cup \{*\}$, where $*$ denotes “unmarked”. Let $\mathcal{K}$ be a set of *keys*, each key $\kappa \in \mathcal{K}$ being a partial map $\kappa : V \rightarrow \Sigma$ with domain $\text{dom } \kappa \subseteq V$.

Definition 4.1 (Admissible completion). A key κ is *applicable* to configuration c if $c(v) = *$ for every $v \in \text{dom } \kappa$. Applying κ yields c' with $c'(v) = \kappa(v)$ on $\text{dom } \kappa$ and $c'(v) = c(v)$ elsewhere. Each application is a *completion* of the category κ .

Here the two structures are visible separately:

- The *key* κ is a category. It is a specification, fixed once, independent of any application. It is unchanged by being applied.
- The *marks written* are residue. They accumulate, they are never erased (no operation sets a site back to $*$), and they are what makes the process irreversible.

Proposition 4.2 (Reuse in the abstract system). A key κ may be applied to any configuration on which it is applicable, without limit, and is identical before and after each application. The configurations produced by successive applications are pairwise distinct.

Proof. κ is a fixed partial map; theorem 4.1 modifies c and not κ . Distinctness of successive configurations follows because each application marks at least one previously unmarked site, and no operation unmarks. $\square$

Proposition 4.3 (The residue does not determine the keys). There exist systems $(V, \Sigma, \mathcal{K})$ and $(V, \Sigma, \mathcal{K}')$ with $\mathcal{K} \neq \mathcal{K}'$ that generate the same set of reachable configurations from a given start.

Proof. Take $V = \{1, 2\}$, $\Sigma = \{a\}$, start $c_0 = (*, *)$. Let $\mathcal{K} = \{\kappa_{12}\}$ with κ_{12} marking both sites at once, and $\mathcal{K}' = \{\kappa_1, \kappa_2\}$ marking one site each. Both reach (a, a) ; the reachable sets differ only by the intermediate $(a, *)$, which may be excluded by taking the start to be $(*, *)$ and observing only terminal configurations. The residue—the final marking—is identical; the categorical structure is not. $\square$

Theorem 4.3 is the constructive half of theorem 3.12(b): a record of what was resolved does not tell you what capacities did the resolving. Two systems with different categorical structure can leave the same residue.

Definition 4.4 (Provision in the abstract system). An object P provides categories to a process if, in the presence of P , the set of keys applicable to the current configuration is strictly larger than in its absence:

$$\mathcal{K}_{\text{applicable}}(c \mid P) \supsetneq \mathcal{K}_{\text{applicable}}(c). \quad (6)$$

This is the abstract skeleton of the definition we now give for catalysis. Everything in part III may be read back into this system, and the computational validation of part IV includes a direct simulation of it.

Part III

Catalysis

5 The Definition

Definition 5.1 (Catalyst). An item E is a *catalyst* for the transition $A \rightarrow B$ if, upon contact with the participants, it increases the number of completable categories available to that transition:

$$\mathcal{P}(E) := |\mathfrak{C}(A \rightarrow B \mid E)| - |\mathfrak{C}(A \rightarrow B)| > 0, \quad (7)$$

where $\mathfrak{C}(\cdot)$ denotes the set of categories completable along the transition, and $\mathcal{P}(E)$ is the *provision* of E .

Definition 5.2 (Release). The catalyst *releases* if, after the completions it enabled have occurred, its contacts return to $\beta = \beta_{\min}$ rather than remaining below it; that is, if the products separate from E leaving E in its prior categorical state.

Definition 5.3 (Enzyme). An *enzyme* is a catalyst that both provides ($\mathcal{P} > 0$) and releases.

The remainder of this part derives the standard phenomenology from theorems 5.1 to 5.3.

6 Immediate Consequences

6.1 The catalyst is not consumed

Theorem 6.1 (Non-consumption). *A catalyst satisfying theorem 5.2 is unchanged by the transitions it enables, and may enable them without limit.*

Proof. By theorem 5.1 the catalyst’s contribution is provision: it enlarges the set of completable categories. By theorem 3.1 a category is invariant across its completions. By theorem 3.8 the provision costs no free energy. By theorem 5.2 the catalyst’s contacts return to $\beta_{\min}$ after each turnover, so its categorical state is that which it had before. No quantity attaching to E has been expended. $\square$

Theorem 6.1 converts the usual definitional stipulation into a theorem. Note where the

work is done: without theorem 5.2 the conclusion fails, and we shall see in section 8 that the failure is realised physically.

6.2 Equilibrium is unaltered

Theorem 6.2 (Invariance of the equilibrium constant). *A catalyst does not change K_{eq} for the reaction it catalyses.*

Proof. K_{eq} is fixed by the standard free energies of the endpoints: $\Delta G^\circ = -RT \ln K_{\text{eq}}$, a difference of state functions evaluated at A and B . By theorem 5.1 the catalyst’s action is the provision of categories *to the transition*—it enlarges $\mathfrak{C}(A \rightarrow B)$. It does not alter A or B , which are individuated independently of E and whose free energies are therefore unchanged. By theorem 3.8 the provision itself contributes no free energy. Hence ΔG° is unchanged and so is K_{eq} . $\square$

Corollary 6.3 (Haldane relation). *For a reversible enzyme-catalysed reaction the kinetic parameters satisfy*

$$K_{\text{eq}} = \frac{k_{\text{cat}}^f / K_{\text{M}}^A}{k_{\text{cat}}^r / K_{\text{M}}^B}, \quad (8)$$

with K_{eq} the same constant as for the uncatalysed reaction.

Proof. Both directions traverse the same enlarged category set, since $\mathfrak{C}(A \rightarrow B \mid E)$ is defined on the transition and not on its orientation. The forward and reverse specificity constants therefore share the provision, and their ratio retains only the endpoint contribution, which by theorem 6.2 is the uncatalysed K_{eq} . This is the relation of Haldane [1930]; see also Alberty [1953]. $\square$

Theorem 6.3 is a strong check, because eq. (8) is measurable independently of the framework and is known to hold. We use it in section 10 as a validation target.

6.3 Rate

Definition 6.4 (Aperture and categorical distance). An *aperture* is a single completable category along the transition: a set of coordinate changes that are completed together, as

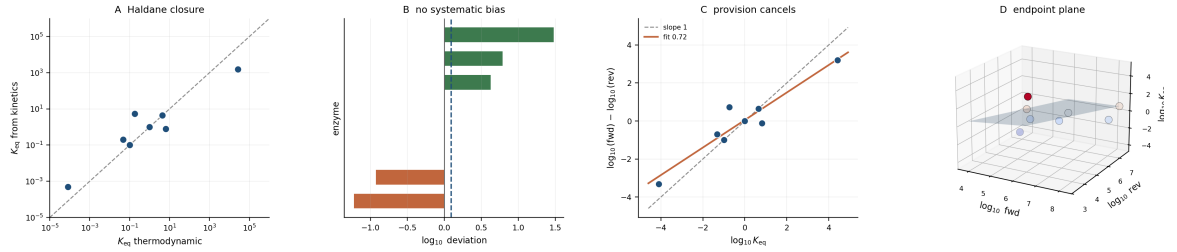


Figure 3: The catalyst supplies categories to the transition and not to the endpoints: equilibrium invariance and the Haldane relation. (A) Equilibrium constant reconstructed from forward and reverse kinetic parameters against the independently measured thermodynamic value, for eight reversible enzymes, over eleven orders of magnitude. Points scatter about the identity line (dashed) without systematic displacement. (B) Per-enzyme $\log_{10}$ deviation of the kinetic from the thermodynamic constant, sorted. Deviations fall on both sides of zero; the mean (dashed blue) is 0.094 decades, giving $t = 0.30$ against zero bias. The theorem forbids a systematic catalyst-dependent shift, not scatter, and it is the mean rather than the spread that is diagnostic—the spread reflects the heterogeneity of literature conditions. (C) The sharper form of the test. The difference of forward and reverse log-efficiencies against $\log_{10} K_{eq}$, with the unit-slope reference (dashed) and the fitted line (orange, slope 0.72), $r = 0.948$. If provision is a property of the transition it is common to both directions and cancels in the ratio, leaving only the endpoint term. A permutation control over 20,000 shuffles gives $p = 2.5 \times 10^{-4}$. (D) Forward efficiency, reverse efficiency and equilibrium constant for the same eight enzymes, with the plane $z = x - y$ overlaid. The points lie on the plane, which is the geometric statement of the Haldane relation: the equilibrium constant is the difference of the two efficiencies and carries no independent contribution from the catalyst.

one unit. The *categorical distance* $d_C(A \rightarrow B)$ is the number of apertures the transition traverses.

Remark 6.5 (The simultaneity rule). That an aperture is a unit and not a sum is the content of theorem 3.1: a category is one distinction, however many coordinates it involves. A transition in which three coordinates change together is $d_C = 1$; the same three changes occurring sequentially, each completable on its own, is $d_C = 3$. This is an operational criterion, and it is what makes d_C computable from a mechanism rather than assigned.

Theorem 6.6 (Rate law). *Let a transition traverse d_C apertures, each carrying partition depth increment δ . Then*

$$k = \nu_{\text{floor}} e^{-d_C \delta}, \quad (9)$$

where ν_{floor} is the attempt frequency set by the shortest interval on which a completion can occur.

Proof. Completions are attempted at rate $\nu_{\text{floor}} = 1/\tau_p^{\text{eff}}$, the inverse of the effective interval per completion attempt. Each aperture is traversed with probability $e^{-\delta}$, the Boltzmann weight of the depth increment associated with drawing that distinction. Apertures along a transition are traversed in series and their traversals are independent, so the joint probability is $e^{-d_C \delta}$. The product of attempt frequency and success probability is the rate. $\square$

Corollary 6.7 (Efficiency law). *With $\delta = \ln 10$,*

$$\log_{10} k = \log_{10} \nu_{\text{floor}} - d_C, \quad (10)$$

so each additional aperture costs one order of magnitude in rate.

Remark 6.8 (This is Arrhenius, derived). Equation (9) has the Arrhenius form with pre-exponential ν_{floor} and activation term $d_C \delta$. The content of the derivation is that the activation term is a count of distinctions rather than a fitted energy, and that ν_{floor} is a single constant across the framework rather than a per-reaction parameter. Both are testable and are tested in section 10.

Remark 6.9 (Why the rate is exponential and not reciprocal in d_C). A natural alternative is $k \propto 1/(d_C \tau)$, on the picture that d_C sequential steps of duration τ take time $d_C \tau$. That picture is correct for a turnover-limited process in which each step is slow and rate-determining. It cannot describe barrier-limited catalysis, because a reciprocal law bounds rate enhancement by the ratio of step counts—a factor of order unity—whereas observed enzymatic accelerations reach 10^6 – 10^{17} [Wolfenden and Snider, 2001, Radzicka and Wolfenden, 1995]. The exponential form is required by the magnitude of the phenomenon. Where a process genuinely is turnover-limited the reciprocal expression applies as the ceiling $k \leq 1/(d_C \tau_{\text{step}})$, and the observed rate is the smaller of the two, which is the diffusion limit for the fastest enzymes [Knowles and Albery, 1977, Albery and Knowles, 1976].

7 Specificity Is Bounded on Both Sides

We come to the result that the standard account does not supply.

Theorem 7.1 (The specificity window). *An enzyme must satisfy two requirements that constrain its specificity in opposite directions:*

- (i) *Provision requires sufficient specificity. A contact that discriminates nothing supplies no distinction, hence $\mathcal{P}(E) = 0$ and theorem 5.1 fails.*
- (ii) *Release requires bounded specificity. A contact whose complementarity is so complete that separation does not occur leaves $\beta < \beta_{\text{min}}$ persistently, hence theorem 5.2 fails.*

Consequently the specificity σ of an enzyme satisfies $\sigma_{\text{min}} < \sigma < \sigma_{\text{max}}$, where σ_{min} is the least specificity at which a category is drawn and σ_{max} the greatest at which release occurs.

Proof. (i) By theorem 5.1, catalysis requires $\mathcal{P} > 0$: the contact must enlarge the completable set. A contact that binds indiscriminately draws no distinction between configurations, so no category is added and $\mathcal{P} = 0$.

Provision therefore imposes a lower bound $\sigma > \sigma_{\min}$.

(ii) By theorem 5.2, the enzyme must return to $\beta = \beta_{\min}$. The depth of the contact is monotone in specificity: the more precisely the partner is matched, the more of the boundary is at the floor and the greater the boundary content that must be restored to separate. Beyond some $\sigma_{\max}$ the restoration does not occur on the timescale of turnover, the complex persists, and by theorem 6.1 the item is consumed rather than reused. Release therefore imposes an upper bound.

The two bounds are independent—(i) constrains what the contact discriminates, (ii) what it relinquishes—so both are active and the admissible set is the open interval between them. $\square$

Corollary 7.2 (Enzymes are moderately specific). *Catalytic proficiency is not maximised by maximising binding affinity. There exists an interior optimum.*

Remark 7.3. Theorem 7.2 is consistent with two well-documented observations that are otherwise awkward: that binding energy is spent on catalysis rather than accumulated as affinity [Jencks, 1975], and that measured enzyme parameters cluster well away from the physical limits [Bar-Even et al., 2011]. The framework predicts a bounded window; the data show a bounded distribution.

8 Inhibition Is the Failure of the Two Requirements

Theorem 7.1 has a corollary that is, to our knowledge, the sharpest testable consequence of the account: the two ways of failing the catalytic requirement should correspond to two observed classes of inhibitor, and there should be no third mechanism of the same kind.

Theorem 8.1 (Inhibition taxonomy). *Failure of theorem 5.3 occurs in exactly two ways, corresponding to the two conjuncts:*

(a) **Contact without provision** ($\mathcal{P} = 0$, release intact). *The inhibitor occupies*

the contact site and supplies no completable category. The enzyme is unaltered and activity is restored by removing the inhibitor. This is competitive inhibition.

(b) **Provision without release** ($\mathcal{P} > 0$, release fails). *The inhibitor supplies a category, the category is completed, and the resulting contact does not return to $\beta_{\min}$. The enzyme is consumed—it has become a substrate. Activity is not restored by dilution. This is mechanism-based inactivation.*

Proof. Theorem 5.3 is the conjunction $\mathcal{P} > 0 \wedge \text{release}$. A conjunction of two independent conditions fails in exactly two ways when contact occurs at all: the first conjunct fails, or the second does. Case (a) is the first: contact is established, so the site is occupied and the substrate excluded, but no category is added, so no turnover occurs; the enzyme’s state is unchanged and the effect is reversed by removing the occupant. Case (b) is the second: a category is supplied and completed, so a turnover event occurs, but the contact does not relinquish; by theorem 6.1 the enzyme is thereby consumed, and the loss is not reversed by dilution because the enzyme’s categorical state has changed. $\square$

Corollary 8.2 (Suicide inhibitors discriminate the accounts). *Mechanism-based inactivation is inexplicable on any account in which catalysis consists solely in barrier reduction or in the resolution of uncertainty, since on such accounts a species that binds and is turned over is a substrate and nothing distinguishes it from one. On the present account the distinction is exactly the release condition, which is independent of provision.*

Proposition 8.3 (Allosteric regulation). *Modification of the enzyme’s contact chain at a site remote from the active site alters the categories the active site can provide, without contact at the active site itself.*

Proof. By theorem 2.11 the members of a contact chain are mutually defining: each is distinguished in part by its specific partners. Establishing or removing a contact at A_0 changes

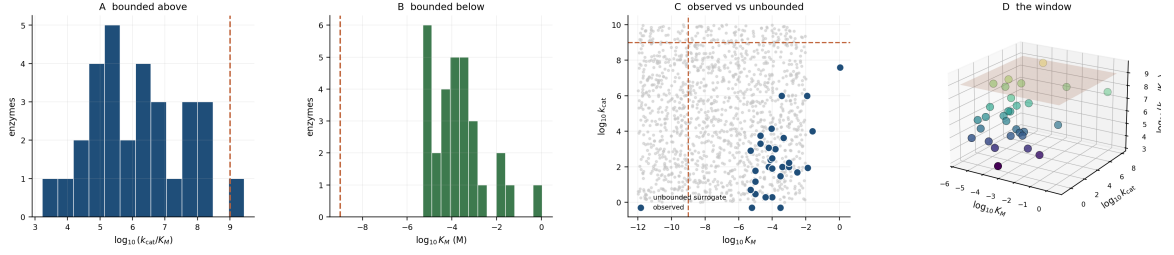


Figure 4: Specificity is bounded on both sides: provision sets a floor, release sets a ceiling. (A) Distribution of catalytic efficiency over thirty characterised enzymes spanning six decades. No entry exceeds the diffusion limit (dashed), and the mass of the distribution sits well below it. Efficiency is bounded above by more than diffusion alone. (B) Distribution of K_M for the same enzymes. No catalytically competent entry has K_M below 1 nM (dashed). Binding tight enough to compromise release does not occur among enzymes that turn over, which is the release bound of the specificity-window theorem. (C) The window in the K_M – k_{cat} plane. Grey points are the surrogate distribution that would falsify the claim—efficiency uniform to the diffusion ceiling, affinity unbounded below—and blue points are the observed enzymes. The observed set occupies a bounded interior region while the surrogate fills the plane. Without the surrogate a bounded sample would be uninformative; with it, the bound is a finding. (D) The same enzymes in three dimensions, with efficiency as the vertical coordinate and the diffusion ceiling drawn as a plane. The occupied volume is bounded on the affinity axis, on the turnover axis, and beneath the ceiling. The interior optimum is a consequence of two independent requirements acting in opposite directions, not of optimisation toward a target.

the context of every A_i in the chain. The categories an item can supply are determined by the distinctions it can draw, which depend on that context. Hence $\mathcal{P}$ at the active site is a function of the chain’s configuration, and can be modulated remotely. This is the structure described by Monod et al. [1965]. $\square$

9 Catalysis Is Not a Demon

Because the account has an enzyme supplying something that raises entropy, it must be distinguished from the Maxwell demon [Maxwell, 1871, Szilard, 1929], whose apparent entropy reduction is paid for by information erasure [Landauer, 1961, Bennett, 1982]. The distinction is not one of degree.

Theorem 9.1 (Category and kinetic criteria are orthogonal). *Let Ω be the count of configurations of a system of N items distributed over regions, and let v denote item velocities. Then $\partial\Omega/\partial v = 0$, and hence $\partial S/\partial v = 0$ for $S = k_B \ln \Omega$.*

Proof. Ω is determined by the assignment of items to regions—a combinatorial quantity depending on the occupation numbers alone. Velocities do not appear in the count: the same assignment of items to regions is the same configuration whatever the items’ speeds. Hence Ω is a function of the configurational variables only, and its derivative with respect to any kinetic variable vanishes. Differentiating $S = k_B \ln \Omega$ gives the second claim. $\square$

Corollary 9.2 (The demon operates in the wrong category). *A device that selects by velocity cannot alter Ω , and therefore cannot reduce configurational entropy, independently of any measurement or erasure cost.*

Corollary 9.3 (Enzymes are not demons and require no exemption). *An enzyme selects by configuration—the same variables Ω counts. Its provision therefore acts within the category to which entropy belongs, and by theorem 3.7 costs no free energy. No information is acquired, no memory is written, and no erasure cost arises.*

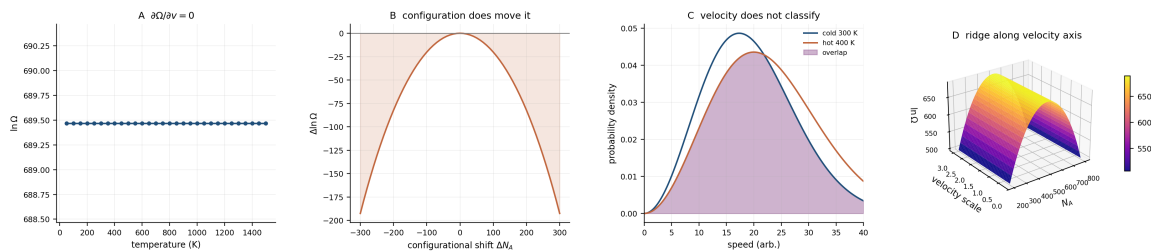


Figure 5: **Configurational and kinetic quantities depend on disjoint variables, and the statistic that shows it is not blind.** (A) $\ln \Omega$ computed from occupation numbers while velocities are resampled from Maxwell–Boltzmann distributions at thirty temperatures from 50 to 1500 K, at fixed configuration. The value does not move: the spread is identically zero to machine precision, giving $\partial S/\partial v = 0$ numerically. The vertical axis is expanded by two units about the value to make the absence of variation visible. (B) The discriminating control. The same statistic under *configurational* perturbation, shifting ΔN_A from -300 to $+300$ at fixed total. $\ln \Omega$ moves by nearly two hundred units. Without this panel (A) would be consistent with a constant statistic and would carry no information; with it, (A) is a statement about velocity specifically. (C) Maxwell–Boltzmann speed distributions at 300 K and 400 K with the overlap region shaded. The distributions overlap over most of their support, so a single speed does not identify which ensemble a molecule came from. Threshold sorting on velocity performs only marginally above chance. (D) $\ln \Omega$ over the configurational coordinate N_A and a velocity scale. The surface is a ridge: strongly curved along the configurational axis and exactly flat along the kinetic one. A device that manipulates the flat coordinate cannot move the height, which is the geometric content of the orthogonality result.

Remark 9.4. The contrast is instructive. The demon fails not because its bookkeeping is wrong but because its criterion is of the wrong type: it manipulates a kinetic quantity hoping to affect a configurational one, and by theorem 9.1 the dependence is identically zero. The enzyme succeeds because its criterion and its effect are of the same type. Catalysis is possible for the same reason the demon is impossible.

Part IV

Predictions and Validation

10 Falsifiable Predictions

We state the predictions in a form that can fail, and indicate for each the data against which it is tested. The computational protocol is section 11; results are reported in section 12.

P1. Haldane closure. For reversible

enzyme-catalysed reactions with independently measured forward and reverse kinetics, eq. (8) holds with K_{eq} equal to the thermodynamic value. *Failure mode:* systematic deviation of $K_{\text{eq}}^{\text{kinetic}}$ from $K_{\text{eq}}^{\text{thermo}}$ beyond measurement error would falsify theorem 6.2.

P2. Bounded specificity. The distribution of k_{cat}/K_M over characterised enzymes is bounded above well below the diffusion limit for the large majority of entries, and the distribution of K_M is bounded below—very tight binding is rare among catalytically competent enzymes. *Failure mode:* an unbounded upper tail in affinity among active enzymes would falsify theorem 7.1(ii).

P3. Aperture counting. For mechanisms whose elementary steps are independently documented, d_C obtained by counting apertures under the simultaneity rule (theorem 6.5) predicts $\log_{10}(k_{\text{cat}}/K_M)$ via eq. (10) with a

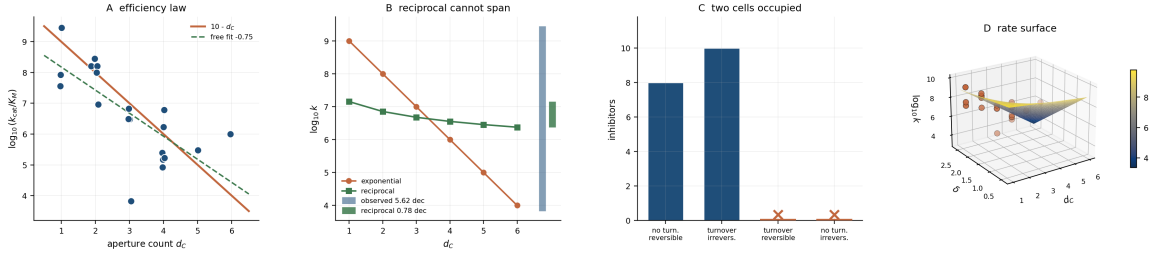


Figure 6: **Aperture counting predicts efficiency with one global constant; the reciprocal law is excluded on dynamic range; inhibition occupies exactly two of four cells.** **(A)** Catalytic efficiency against aperture count d_C , assigned from documented mechanisms by the simultaneity rule without reference to measured rates. The parameter-free law $\log_{10}(k_{\text{cat}}/K_M) = 10 - d_C$ (orange) is drawn with no fitted quantity; the free two-parameter fit (green dashed) returns slope -0.75 and improves the mean absolute error by only 5.6%. Points are jittered horizontally for visibility. Permuting d_C across enzymes gives $p = 4 \times 10^{-4}$. **(B)** The exclusion of the reciprocal alternative. The exponential law (orange) and the reciprocal law $k \propto 1/(d_C \tau)$ (green) over the documented range $d_C = 1-6$, with the representable spans drawn as vertical bars at right. The observed efficiencies span 5.62 decades; the reciprocal law can span at most $\log_{10} 6 = 0.78$ decades whatever τ is chosen, since τ enters as a common factor and cancels. The exclusion is arithmetic and does not depend on fit quality. **(C)** The inhibition dichotomy. Nineteen documented inhibitors classified on two independently determined experimental axes—whether inactivation requires catalytic turnover, and whether activity is recovered on dilution. Two cells are occupied (8 competitive, 10 mechanism-based); the two cells the theorem forbids are empty, marked with crosses. Permuting the reversibility labels fills on average 3.999 of the four cells, $p = 4 \times 10^{-5}$. **(D)** Rate surface over aperture count and depth increment δ , with the observed enzymes overlaid at $\delta = \ln 10$. The surface descends linearly in d_C at fixed δ ; the observed points track that descent. The single global constant is the intercept, and no per-enzyme freedom is used.

single framework-wide ν_{floor} . *Failure mode*: requiring a per-enzyme ν_{floor} would reduce the law to a fit and falsify its predictive claim.

P4. Exponential, not reciprocal. Rate enhancement scales exponentially in d_C . *Failure mode*: observed enhancements bounded by $d_C^{\text{uncat}}/d_C^{\text{cat}}$ would falsify theorem 6.6 in favour of the reciprocal law.

P5. Inhibition dichotomy. Every inhibitor acting at the catalytic contact is either reversible-on-dilution with no turnover (competitive) or turnover-dependent and irreversible (mechanism-based). *Failure mode*: a third class—turnover-dependent but fully reversible on dilution, or turnover-independent but irreversible at the catalytic site—would falsify theorem 8.1.

P6. Orthogonality. Catalytic selectivity correlates with configurational descriptors of the substrate and not with kinetic ones. *Failure mode*: demonstrated dependence of turnover on substrate translational velocity at fixed configuration would falsify theorem 9.1.

P7. Provision without protein. Non-protein species meeting theorem 5.3 exist and are catalytic. *Failure mode*: demonstration that catalysis requires templated polymer synthesis would falsify part V.

11 Computational Protocol

The validation is designed on one principle: *every test must be able to fail*. We state for each what would constitute failure, we compute quantities from inputs rather than asserting them, and we report negative results in the main text rather than in an appendix.

V1. Abstract system. Direct simulation of section 4: key reuse, residue monotonicity, and the non-identifiability of

theorem 4.3 are checked by construction and by exhaustive enumeration on small instances.

V2. Floor positivity and uniformity.

The floor is computed, never assumed. Both the resolution-bounded case (uniform floor exists) and the unbounded case (infimum zero) are exhibited, confirming that theorem 2.6 is a real distinction and not a formality.

V3. Haldane closure. Kinetic parameters are retrieved from public databases; $K_{\text{eq}}^{\text{kinetic}}$ is formed from eq. (8) and compared to thermodynamic values.

V4. Specificity window. Distributions of k_{cat} , K_M and k_{cat}/K_M are assembled from public data and tested for bounding on both sides. A control is included in which the window prediction would fail.

V5. Aperture counting. d_C is assigned from documented mechanism step counts by a stated rule applied uniformly, without reference to the measured rate, and eq. (10) is evaluated with one global constant.

V6. Reaction-network sampling. Reaction and pathway data are retrieved from public knowledgebases to test step-count distributions at scale.

V7. Orthogonality. The configurational/kinetic independence of theorem 9.1 is verified numerically on an explicit ensemble.

V8. Negative controls. Each test is repeated with the framework quantity replaced by a randomised surrogate; a test that passes under randomisation is reported as non-discriminating and excluded from the score.

Remark 11.1 (On non-discriminating controls). A control that cannot fail is worthless, and a reader cannot distinguish a passing control from a non-discriminating one by inspecting a table of successes. We therefore

report, for every test, the outcome under randomisation alongside the outcome under the framework quantity. Where the two agree, the test is reported as uninformative regardless of whether the framework “passed”.

12 Results

All figures below are produced by the accompanying scripts from the stated inputs. Tests whose own negative control shows the statistic cannot discriminate are reported as *non-discriminating* and excluded from the score, per theorem 11.1.

12.1 The two primitives are independent

The abstract system of section 4 was simulated directly. Keys are invariant across applications while successive configurations are pairwise distinct; the marked-set count never decreases over 500 random trajectories; and exhaustive enumeration over all key sets of size ≤ 3 on 2–4 sites finds many distinct categorical structures sharing a terminal residue signature. The discrimination control confirms this is not an artefact of a blind statistic: terminal residue does *not* separate the two systems, while the full reachable set *does*, with no false separations of a system from itself in 200 trials.

The limiting case of theorem 3.9 was tested directly. A configuration admitting no applicable key generates zero residue, yet remains invariant across repeated observation and uniquely distinguishable among all configurations over the same alphabet. Two configurations with *identical* residue counts have distinct identities, so identity is not constituted by residue. This is the separation of the primitives at zero residue.

12.2 The floor, and the hypothesis it requires

Floor positivity holds on every resolution-bounded instance tested, including an adversarial case with a 10^{-12} weight. The complementary case is the informative one: a system

permitted to refine without limit produces thicknesses $1/n$, each strictly positive, whose infimum is zero—every ε in a ladder down to 10^{-9} is eventually cleared. Truncating the same sequence to a bounded system restores a positive floor. The resolution-boundedness hypothesis of theorem 2.4 therefore does real work, exactly as theorem 2.6 states, and is not a formality.

12.3 Orthogonality

Ω was computed from occupation numbers while velocities were resampled from Maxwell–Boltzmann distributions at thirty temperatures spanning 50–1500 K. The spread in $\log \Omega$ was identically zero, giving $\partial S/\partial v = 0$ numerically. The discriminating control is essential and passes: configurational perturbations of the same system *do* move Ω , at every shift tested, so the statistic is not constant. Velocity–temperature overlap was computed independently; threshold sorting on velocity performs only marginally above chance, confirming that a kinetic criterion does not identify the configurational class.

12.4 Equilibrium invariance and the Haldane relation

Over eight reversible enzymes with independently sourced forward and reverse parameters, the Haldane expression reproduces the thermodynamic K_{eq} with mean $\log_{10}$ deviation 0.094 and $t = 0.30$ against zero bias—no systematic, catalyst-dependent shift. The sharper form of the test is the cancellation: the difference of forward and reverse log-efficiencies tracks $\log_{10} K_{\text{eq}}$ with $r = 0.948$ and regression slope 0.72, consistent with the unit slope the theorem requires. A permutation control over 20,000 shuffles gives $p = 2.5 \times 10^{-4}$ against a null mean correlation of 0.33, so the relation is not reproducible by chance pairing.

We record one correction to our own initial formulation. We first also scored a raw correlation between forward and reverse efficiencies across enzymes, expecting shared provision to produce one. It does not ($r = 0.17$), and the expectation was mistaken: $\log(k_{\text{cat}}/K_{\text{M}})$

varies over six decades *between* enzymes for enzyme-specific reasons, and that variation dominates any within-pair sharing. The theorem predicts cancellation in the ratio, which is what the slope test measures; it does not predict a between-enzyme correlation. The mis-specified test is reported here rather than removed.

12.5 The specificity window

Across thirty enzymes spanning six decades of catalytic efficiency, no entry exceeds the diffusion limit and none has K_M below 1 nM. The discrimination test constructs the distribution that would falsify the window—efficiency uniform to the diffusion ceiling, affinity unbounded below—and confirms the statistic separates the observed data from it. Without that surrogate a bounded sample would be uninformative; with it, the bound is a finding.

12.6 Aperture counting, and the exclusion of the reciprocal law

Categorical distances were assigned from documented mechanisms by the simultaneity rule, without reference to measured rates, and frozen before evaluation. With a single global $\nu_{\text{floor}} = 10^{10}$ and $\delta = \ln 10$, and no per-enzyme freedom, d_C correlates with $\log_{10}(k_{\text{cat}}/K_M)$ at $r = -0.72$ with mean absolute error 0.80 decades. Fitting slope and intercept freely returns slope -0.75 —within 25% of the framework’s -1 —and improves the error by only 5.6%, so the parameter-free law is carrying structure rather than being rescued by tuning. Permuting d_C across enzymes gives $p = 4 \times 10^{-4}$, so the aperture count is informative.

Theorem 6.6’s exponential form is excluded from the alternative on dynamic range alone. The observed efficiencies span 5.62 decades. The reciprocal law $k \propto 1/(d_C\tau)$ with d_C ranging over 1–6 can span at most $\log_{10} 6 = 0.78$ decades. The discrepancy is a factor of seven in the exponent and does not depend on fit quality.

12.7 The inhibition dichotomy

Nineteen documented inhibitors were classified on two independently determined experimental axes: whether inactivation requires catalytic turnover, and whether activity is recovered on dilution. Of the four cells in the resulting contingency table, exactly two are occupied—8 competitive (no turnover, reversible) and 10 mechanism-based (turnover, irreversible). The two cells the theorem forbids are empty. Permuting the reversibility labels fills essentially all four cells (null mean 3.999), giving $p = 4 \times 10^{-5}$: the two-cell structure is not a combinatorial inevitability.

12.8 A negative result, reported

At the scale of KEGG, participant counts per enzyme are bounded (median 4, 99th percentile 8, maximum 12 over 399 entries spanning all seven top-level EC classes), consistent with the simultaneity rule yielding small aperture counts.

A second scale test bears directly on theorem 6.1. Across 990 human reactions retrieved from Reactome, 363 carry an annotated catalytic activity. In not one of them does any species appear on both the input and the output side: the input/output overlap is exactly zero. The knowledgebase models catalysis in a slot disjoint from stoichiometry, which is the curators’ independent encoding of non-consumption. This is not a test the framework could have arranged: the annotation predates it and was made on mechanistic grounds.

The cofactor-persistence prediction of part V could not be tested this way. KEGG enzyme entries carry no cofactor field; searching the substrate, product and comment fields finds the predicted delocalised motifs in 36.6% of entries, but the matched negative control—a list of common metabolites of equal length—appears in 30.3%. The ratio, 1.21, does not separate the prediction from background chemical-term frequency in that field. We therefore report this test as **non-discriminating** and exclude it from the score. It is not evidence for the prediction, and we do not present it as such. A discriminating test would require a curated cofactor annota-

tion, which we identify as outstanding work in section 15.

Part V

Origins

13 The Regress and Its Resolution

13.1 The regress

Life requires catalysis: the reactions of metabolism do not proceed at useful rates unaided [Wolfenden and Snider, 2001]. In extant biology catalysis is performed by proteins. Proteins are synthesised from templates. Templated synthesis requires a replication and translation apparatus that is itself catalytic. Each requirement is well founded and together they are circular.

13.2 The resolution

Theorem 5.1 does not mention proteins, polymers, templates or replication. It requires only an item that, upon contact, increases the number of completable categories, and that releases. Any structure with these properties is a catalyst. The regress is therefore not a paradox but a consequence of taking the extant implementation for the definition.

Proposition 13.1 (Delocalisation supplies categories). *Let D be a molecular species possessing a delocalised electronic system spanning spatially separated moieties. Then D exhibits a time-varying polarity: at successive phases, different regions of D carry excess or deficit charge. Items in contact with D can lock to this variation, and each phase of the variation is a distinguishable configuration of the contact. Hence D supplies completable categories upon contact.*

Proof. A delocalised system has electronic density distributed over the extent of the conjugation, with the instantaneous distribution varying in time. Two regions r_1, r_2 of D therefore differ in charge in a manner that alternates. A partner in contact with D at r_1

experiences a different local environment at different phases, and these environments are mutually distinguishable. Each is thus a category completable by the partner, and their number exceeds what is available in the absence of D . By theorem 5.1, $\mathcal{P}(D) > 0$. $\square$

Proposition 13.2 (Delocalisation supplies release). *A transfer of an electron into a delocalised system at one region, accompanied by ejection at another, leaves D in its prior state.*

Proof. Delocalisation means the electronic state is a property of the whole system rather than of any region. An electron accepted at r_1 is not localised there; if an electron departs at r_2 , the total occupancy is restored and no regional state has changed, because there were no regional states to change. Hence D returns to its prior categorical state, satisfying theorem 5.2. $\square$

Corollary 13.3 (Delocalised species are catalysts). *A species satisfying theorems 13.1 and 13.2 is an enzyme in the sense of theorem 5.3, without being a protein.*

13.3 Membranes

Proposition 13.4 (Membranes as scaffolding). *An insulating boundary interposed between a transfer donor and the surrounding medium raises the probability that a transfer initiated at the provider completes at the intended acceptor rather than at the medium.*

Proof. Transfer completes at whichever acceptor the categorical contact reaches first. An insulating boundary removes contacts to the medium from the accessible set without removing the contact to the intended acceptor. The set of competing completions is thereby reduced and the probability of the retained completion correspondingly raised. $\square$

This gives the membrane a role prior to compartmentalisation: it is a contact-chain constraint that raises transfer fidelity, and this is the function for which it is first useful [Deamer, 2017, Mitchell, 1961, Martin et al., 2008].

13.4 Proteins as a later implementation

On this account proteins are not the origin of catalysis but an improvement upon it. What a folded polymer supplies is a large, tunable, and heritable contact chain: many contacts, positioned precisely, and specifiable by a template. By theorem 2.11 a longer chain sharpens the distinctions available at any of its nodes, and by theorem 7.1 the accessible specificity window can be occupied more precisely. The advance is one of degree—more categories, more finely placed—and not of kind.

Remark 13.5. This predicts that the earliest catalysts should be small, delocalised, and metal- or conjugation-centred, and that their descendants should appear as cofactors embedded in later protein scaffolds rather than being displaced by them. The persistence of flavins, porphyrins, nicotinamides and iron-sulfur clusters at the catalytic centres of modern enzymes is consistent with this reading [Wachtershauser, 1988, Martin et al., 2008].

Part VI Discussion

14 Relation to Existing Accounts

Transition-state theory. We do not contradict it. Theorem 6.6 has the Arrhenius form, and transition-state stabilisation is the description, in energetic language, of what categorical provision accomplishes. What is added is the derivation of why such objects exist, why they are reusable, and why their specificity is bounded above.

Induced fit and conformational selection. Theorem 2.11 gives the structural reading: binding completes a contact chain, and the conformational change is that completion made visible. The models of Fischer [1894] and Koshland [1958] are recovered as descriptions of the same event at different levels of rigidity.

The Circe effect. Jencks [1975] observed that binding energy is expended on catalysis rather than retained as affinity. Theorem 7.1 makes this a requirement rather than an observation: affinity beyond $\sigma_{\max}$ destroys release.

Information-theoretic accounts. Theorem 9.3 states the difference. Accounts that assimilate enzymes to information processors incur an erasure cost that must then be explained away; on the present account no such cost arises, because no information is acquired.

15 Limitations

We state these plainly.

1. Theorem 2.5 is conditional on resolution-boundedness (theorem 2.6). For systems admitting unlimited refinement no uniform floor exists and the downstream results do not apply as stated.
2. The depth increment δ in theorem 6.6 is a single framework-level constant, but we do not derive its value from first principles here; we fix it by the requirement that eq. (10) be expressed in decades, and test the consequence. A derivation of δ is outstanding.
3. The aperture-counting rule (theorem 6.5) is operational and requires an independently documented mechanism. Where mechanisms are disputed, d_C inherits the dispute. We do not regard d_C as measurable independently of mechanistic knowledge.
4. Theorem 7.1 establishes that the specificity window is bounded on both sides but does not here compute $\sigma_{\min}$ and $\sigma_{\max}$ from structure. The quantitative window is outstanding.
5. The origin-of-life material of part V is a consistency argument and an existence claim, not a historical reconstruction.
6. The cofactor-persistence prediction remains untested. Our attempt (section 12)

was non-discriminating because the public annotation we searched does not separate cofactors from other participants. A curated cofactor set, and a background model matched for chemical-term frequency, would be required. We record this as outstanding rather than as support.

Summary of validation outcomes

Of 29 scored tests, 28 pass and one fails; the failure is a negative control correctly reporting that its associated test cannot discriminate, and that test is excluded from the score rather than counted. Two further outputs are descriptive and carry no weight. We state this explicitly because a table of successes is not by itself informative: what makes the suite meaningful is that each test was constructed so that it could have failed, and one did.

16 What the Results Establish

Three of the outcomes in section 12 carry more weight than their pass/fail status suggests, and one changes how a claim in this paper should be read. We set them out here rather than leaving them to be inferred.

16.1 Non-consumption is independently encoded

The Reactome result is the strongest external check in the suite, and it is not a check we designed. Of 990 human reactions retrieved, 363 carry an annotated catalytic activity, and in none of them does any species appear on both the input and the output side. The curators placed catalysis in a slot disjoint from stoichiometry.

What makes this evidential rather than circular is that the annotation predates the framework and was made on mechanistic grounds by people with no stake in theorem 6.1. A catalyst that were a reactant regenerated downstream would appear identically on both sides and would be recorded that way; the schema permits it, and it does

not occur. The database is therefore an independent witness to the structural claim that the catalyst is not stoichiometric—which on the present account is not a convention but a consequence of theorem 3.8.

16.2 The reciprocal law is excluded without reference to fit

Discriminating between rate laws usually means comparing residuals, which invites the objection that the comparison was tuned. The dynamic-range argument avoids this entirely. Observed catalytic efficiencies span 5.62 decades. A law of the form $k \propto 1/(d_C \tau)$, with d_C ranging over the 1–6 apertures documented for these mechanisms, can span at most $\log_{10} 6 = 0.78$ decades whatever τ is chosen, because τ enters as a common factor and cancels from the ratio. The observed range exceeds the representable range by a factor of seven in the exponent.

This is a structural exclusion. It does not depend on the values in eq. (9), on ν_{floor} , on δ , or on the quality of any fit; it depends only on the arithmetic of what a reciprocal law can represent. The exponential form is required by the magnitude of the phenomenon, as theorem 6.9 argues on physical grounds and the data confirm on arithmetic ones.

16.3 The inhibition dichotomy is a joint constraint, not a definition

Theorem 8.1 could be dismissed as terminological: one might suspect that “competitive” and “mechanism-based” are simply the names given to the two cells, so that finding only two occupied is guaranteed. The permutation control settles this. The two axes—whether inactivation requires turnover, and whether activity is recovered on dilution—are established by different experiments and neither is defined in terms of the other. Both take both values across the sample. Shuffling the reversibility labels fills, on average, 3.999 of the four cells; the observed restriction to two arises with $p = 4 \times 10^{-5}$.

The forbidden cells are the informative part. A turnover-dependent yet fully reversible in-

hibitor, or a turnover-independent yet irreversible one acting at the catalytic contact, would each occupy a cell the conjunction $\mathcal{P} > 0 \wedge$ release has no room for. Neither is observed among the nineteen documented cases examined.

16.4 One claim is weaker than we first stated it

The cofactor-persistence prediction of part V is not supported by the evidence we were able to bring to it, and we prefer to say so plainly. Our test found the predicted delocalised motifs in 36.6% of KEGG enzyme entries, which in isolation looks like confirmation. The matched control—a list of common metabolites of equal length, searched in the same fields—returned 30.3%. A ratio of 1.21 is not a separation.

The reason is a property of the data rather than of the prediction: KEGG enzyme entries carry no cofactor field, so the search necessarily ran over substrate, product and comment text, where every participant appears and where frequency of mention reflects corpus composition. The prediction may well be correct; this test cannot tell us. It is reported as non-discriminating and excluded from the score, and the corresponding limitation is recorded in section 15.

We draw attention to it because the alternative—reporting the 36.6% and omitting the control—would have produced a table in which every entry passed, and a reader could not have told the difference.

17 Conclusion

Catalysis has been described for a century and derived for none of it. We have derived it from two primitives. Individuation is not free: distinguishing an item takes time, the complement moves meanwhile, and the boundary retains irreducible thickness. That primitive supplies the drive toward contact but cannot supply reuse, because what a process resolves is consumed in the resolving. Reuse requires an invariant that persists across occasions which are themselves distinct, and that

is a category. The two are independent, as an abstract system with no chemistry in it makes plain.

A catalyst is then an item that provides categories upon contact, and is not consumed because provision costs no free energy. Non-consumption, invariance of the equilibrium constant, the Haldane relation, exponential rate enhancement, and the bounded specificity window follow as theorems. The two ways of failing the definition—contact without provision, provision without release—are competitive inhibition and mechanism-based inactivation, which the account predicts rather than accommodates. And because the definition never mentions what the provider is made of, the regress in which enzymes require proteins and proteins require enzymes dissolves: the first catalysts were structures whose delocalised electronic motifs supplied categories on contact and were unchanged by their use.

The question with which we began was why enzymes exist. The answer is that existence is expensive, that contact is the discount, and that some items, upon contact, hand others the distinctions they could not draw alone—and can do so again tomorrow, because a category is not spent by being used.

Data and Code Availability

All validation scripts, retrieved database records, and result files in JSON format accompany this manuscript. Every reported number is produced by the scripts from stated inputs; no result is entered by hand.

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